

Tableau for Data Science

Session 1 — Introduction, Installation & Workspace Overview

1. Introduction to Tableau

Real Life Analogy:

Imagine you are an artist  with a huge canvas.
You have raw data (paints of different colors).

-  Without Tableau: You manually arrange numbers in Excel — static, boring
-  With Tableau: You drag & drop data onto canvas — beautiful visuals appear instantly!

Tableau is like giving a paintbrush to your data —
it turns boring numbers into stunning, interactive dashboards effortlessly!

 "Tableau = The world's most powerful drag-and-drop data visualization tool"

What is Tableau?

Tableau is a leading Business Intelligence and Data Visualization tool that allows users to connect to data, create interactive charts/dashboards, and share insights — all without writing code.

Developed by: Tableau Software (acquired by Salesforce in 2019)
Headquarters: Seattle, Washington, USA

 Simple Meaning: "Connect Data → Drag & Drop → Stunning Visuals → Share"

1.1 Why Tableau for Data Science?

As a Data Science student, Tableau helps you:

- Explore and understand datasets visually before modeling
- Present ML model results to non-technical stakeholders
- Build end-to-end data stories with interactive dashboards
- Perform quick EDA (Exploratory Data Analysis) without coding
- Connect to big data sources like Hadoop, Spark, SQL databases

1.2 Tableau Product Family

Tableau is not just one product — it is a family of tools:

 Tableau Desktop Full-featured desktop app for building reports ✓ Individual analysts & developers 💰 Paid — ₹5,500+/mo	 Tableau Public Free desktop app — publish to public web ✓ Students & learners 💰 FREE ✓	 Tableau Online Cloud-hosted Tableau Server ✓ Teams using cloud 💰 Paid — SaaS	 Tableau Server On-premises server for enterprise sharing ✓ Large enterprises 💰 Paid — Enterprise	 Tableau Mobile View dashboards on iOS & Android ✓ Managers on the go 💰 Free (with server)
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💡 Which version will we use in this course?
We will use TABLEAU PUBLIC — it is 100% FREE!
Download from: public.tableau.com
Limitation: Reports are published publicly (no private saving locally)
Perfect for learning — all features of Desktop available!

1.3 Tableau vs Power BI

Feature	Tableau	Power BI
Developer	Salesforce (Tableau Software)	Microsoft
Free Version	✓ Tableau Public (free)	✓ Power BI Desktop (free)
Visualization	★ Industry leading — more chart types	Good — improving rapidly
Ease of Use	Drag & drop — very intuitive	Slightly steeper DAX learning curve
Calculations	Calculated Fields (easy syntax)	DAX (powerful but complex)
Data Capacity	Handles large data well	1 GB limit (Pro) / larger Premium

Integration	Connects to 100+ data sources	Best with Microsoft ecosystem
Collaboration	Tableau Server / Online	Power BI Service (better for teams)
Best For	Data visualization & storytelling	Business reporting & Microsoft stack

★ Which one to learn?

Learn BOTH — companies use both tools!

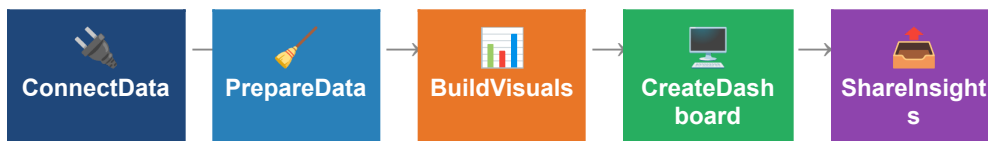
Tableau = Better for visuals & storytelling

Power BI = Better for Microsoft environment & DAX calculations

Having both on your resume makes you stand out! 🚀

1.4 How Tableau Works — The Workflow

Tableau follows a simple 5-step workflow:



Step	Stage	What Happens
1	 Connect Data	Connect to Excel, CSV, SQL, Google Sheets, web etc.
2	 Prepare Data	Clean, join tables, create calculated fields in Data Source tab
3	 Build Visuals	Drag dimensions & measures to Rows/Columns/Marks shelves
4	 Create Dashboard	Combine multiple sheets into one interactive dashboard
5	 Share Insights	Publish to Tableau Public / Server / Online for sharing

2. Installation Guide

 Real Life Analogy:

Installing Tableau Public is just like installing any app:

👉 Go to website → Download → Install → Open → Start exploring!

Best part — Tableau Public is completely FREE 🎉

No credit card, no trial period — just download and use!

2.1 System Requirements

Requirement	Specification
Operating System	Windows 10/11 (64-bit) OR macOS 10.15 (Catalina) or later
Processor	Intel or AMD processor — 1.5 GHz or faster
RAM	Minimum 2 GB Recommended 8 GB or more
Storage	At least 1.5 GB free disk space
Display	1366 × 768 minimum screen resolution
Internet	Required for download, publishing & Tableau Public account
Account	Free Tableau Public account (sign up at public.tableau.com)

2.2 Download & Installation Steps

Follow these steps to download and install Tableau Public:

1. Open browser → go to: <https://public.tableau.com/en-us/s/download>
2. Enter your email address → Click 'Download the App'
3. The installer file downloads (.exe for Windows / .dmg for Mac)

4. Double-click the installer → Accept the License Agreement
5. Click 'Install' → Wait for installation to complete → Click 'Finish'
6. Tableau Public opens automatically → Sign in with your free account



Creating a Free Tableau Public Account:

1. Go to: <https://public.tableau.com>
 2. Click 'Sign Up' → Enter name, email, password
 3. Verify email → Account ready!
- This account lets you save and publish your work online for FREE.

2.3 First Launch — Start Screen

When Tableau Public opens for the first time you will see the Start Screen:

Left Panel	Centre Panel	Right Panel
Connect• To a File (Excel, CSV, PDF...)• To a Server (SQL, Google Sheets...)• Saved Data Sources	OpenRecent workbooks will appear here(Empty on first use)	DiscoverSample workbooks from Tableau Public gallery to inspire your first viz!

Connect

To a File

Microsoft Excel
Text file
JSON file
Microsoft Access
PDF file
Spatial file
Statistical file

To a Server

OData
More... >

Open

Open from Tableau Public

What's New

You can now save your work locally or publish to your Tableau Public profile. Local save is available on Tableau Desktop Public Edition 2026.1.1

UPDATE NOW →

Learn Tableau

 [Get started with Tableau Public](#) [Watch how-to videos](#) [Explore sample data sets](#) [Learn on Trailhead: Tableau Hands-On Challenges](#) [Read the blog](#)[EXPLORE TODAY'S VIZ →](#)[SEE TRENDING VIZZES →](#)[VIZ OF THE DAY ARCHIVE →](#)

3. Workspace Overview

🔍 Real Life Analogy:

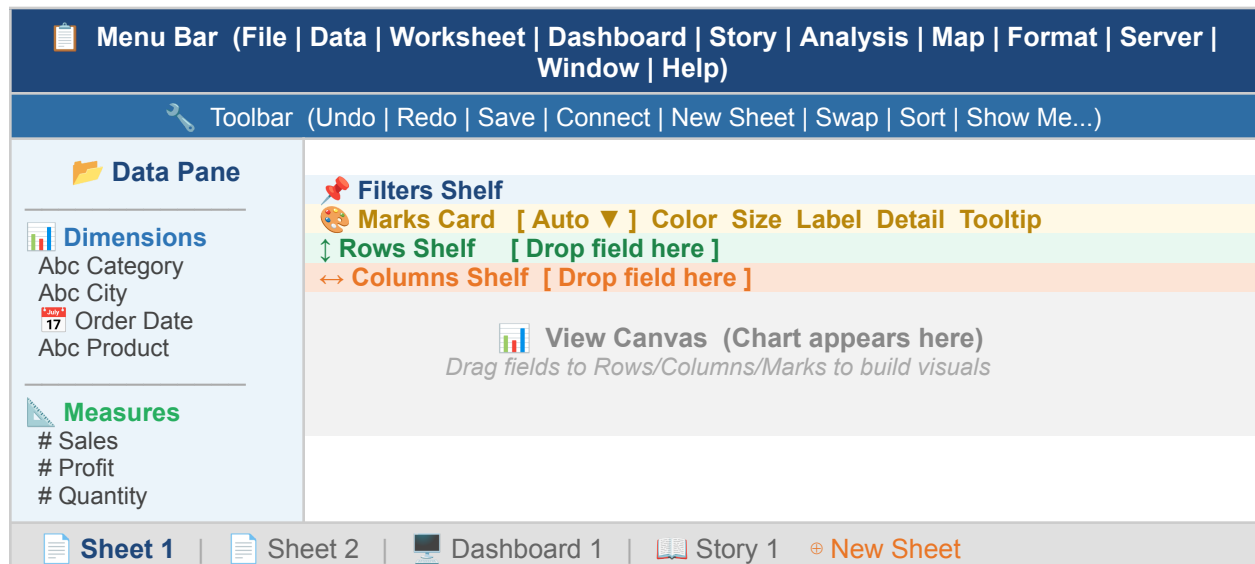
Think of Tableau workspace like a professional kitchen 🧑🍳

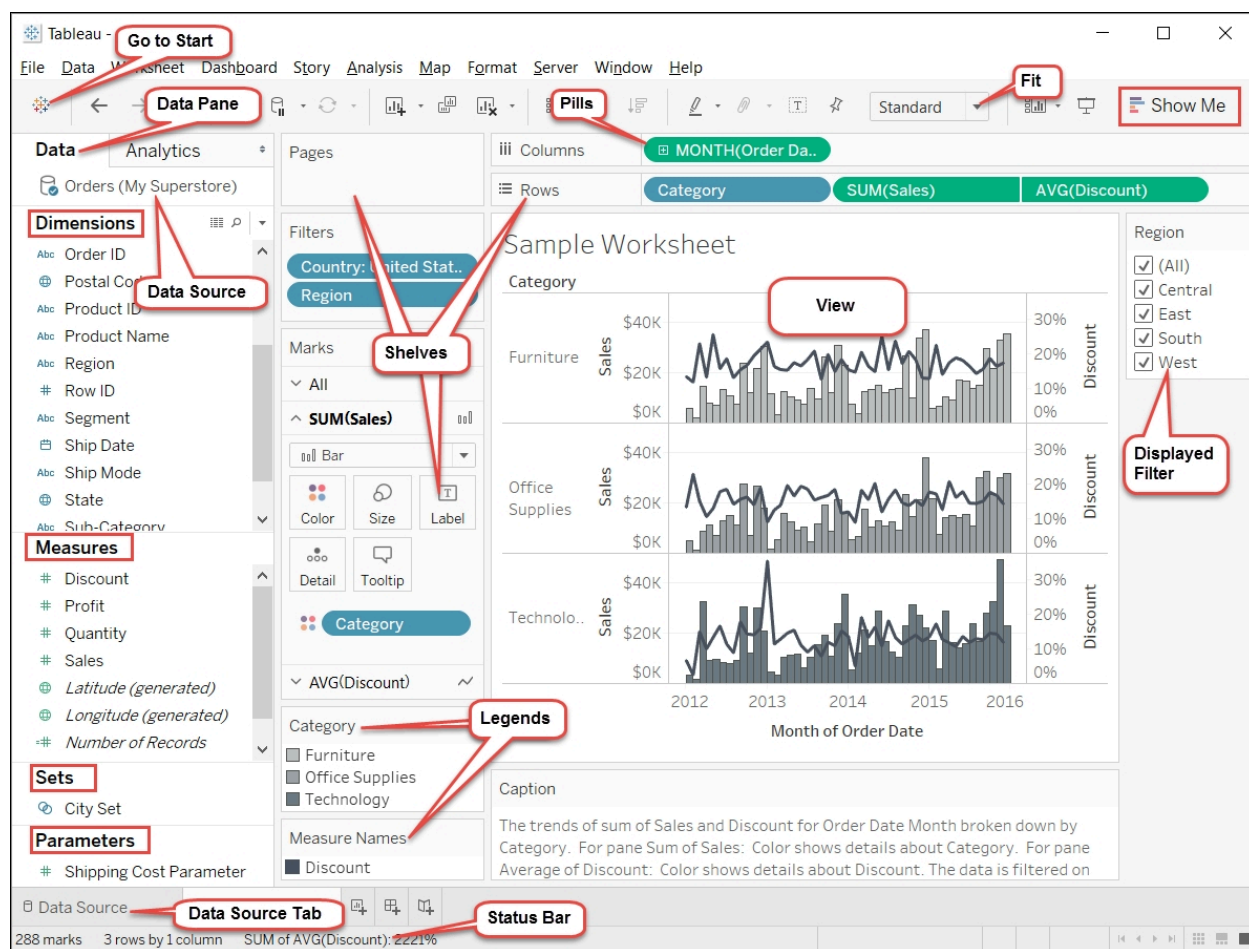
- 📁 Data Pane = Your refrigerator (all ingredients stored here)
- ↔ Columns Shelf = Horizontal chopping board (left to right)
- ↑↓ Rows Shelf = Vertical chopping board (top to bottom)
- 🎨 Marks Card = Your seasoning box (color, size, shape of data)
- 🔍 Filters Shelf = Your sieve (filter out unwanted ingredients)
- 📊 View Canvas = The cooking plate (final dish appears here)
- 📖 Show Me Panel = Recipe book (suggests best chart for your data)

📌 Drag ingredients from fridge → place on boards → dish appears on plate!

3.1 Complete Interface Diagram

Here is the complete layout of Tableau's workspace:





3.2 Menu Bar

The Menu Bar at the top has all commands organized in menus:

Menu	Key Options Inside
File	New, Open, Save, Export as PDF/Image, Print
Data	Connect to new data, Edit data source, Refresh extract
Worksheet	New worksheet, Export, Show summary, Tooltip settings
Dashboard	New dashboard, Device layouts, Actions
Analysis	Aggregation, Forecasting, Trend lines, Reference lines
Map	Map layers, Geocoding, Map options
Format	Font, Borders, Lines, Shading, Animations
Server	Sign in, Publish workbook, Open from Tableau Server

3.3 Data Pane — Your Data Ingredients

The Data Pane is on the left side — it shows all columns from your connected data source:

 Dimensions (Blue)	 Measures (Green)
Categorical / text / date fields	Numeric fields that can be aggregated
e.g. Customer Name, City, Category, Date	e.g. Sales, Profit, Quantity, Discount
Used for grouping & slicing data	Used for calculating totals, averages etc.
Shown with Abc icon (text) or  (date)	Shown with # icon (number)
Drag to Rows/Columns for categories	Drag to Rows/Columns for values

Dimensions vs Measures — Key Rule:

Tableau automatically classifies fields when you connect data:

Text fields → Dimensions (blue) e.g. City, Product, Name

Number fields → Measures (green) e.g. Sales, Profit, Qty

Date fields → Dimensions (blue) with date hierarchy

You can change classification: Right-click any field → Convert to Measure/Dimension

 Wrong classification = Wrong charts! Always verify after connecting data.

3.4 Shelves — Where You Build Your Chart

Shelves are the key building blocks of every Tableau visualization. Drag fields onto shelves to build charts:

 TABLEAU SHELVES — Where You Build Your Chart
↔ COLUMNS SHELF Controls the X-axis (horizontal) of your chart. Drag Dimensions here for categories, Measures for continuous axis. <i>Example: Drag 'Month' here → X-axis shows Jan, Feb, Mar...</i>
↑↓ ROWS SHELF Controls the Y-axis (vertical) of your chart. Drag Measures here for values. <i>Example: Drag 'SUM(Sales)' here → Y-axis shows revenue values.</i>
 FILTERS SHELF

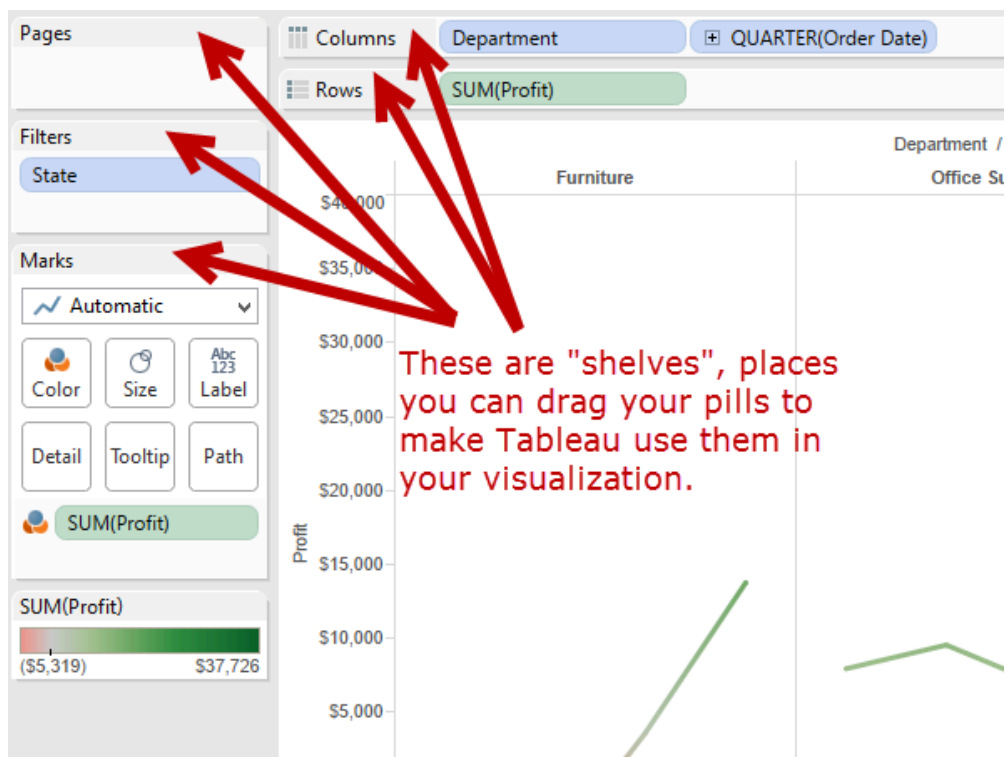
Limit which data is shown in the view. Drag any field here and set conditions.
Example: Drag 'Region' → choose 'West' → only West data shown.



MARKS CARD

Controls the visual properties of marks (data points) in your chart:

- Color → Color marks by a field (e.g. color bars by Category)
- Size → Size marks by a value (e.g. bigger bubbles = higher sales)
- Label → Show text labels on marks
- Tooltip → Data shown on hover
- Shape → Change mark shape (circle, square, star...)



3.5 Show Me Panel

The Show Me panel is on the top right — it suggests the best chart type for your selected fields:



SHOW ME PANEL — Smart Chart Recommender

How it works:

1. Select fields in the Data Pane (Ctrl+Click to select multiple)
2. Look at Show Me panel — recommended charts are highlighted in color
3. Click any chart type → Tableau builds it automatically!

Available chart types in Show Me:

Bar | Line | Pie | Scatter | Map | Treemap | Gantt | Bullet | Histogram | Box Plot | Area | Heat Map | Highlight Table | Symbol Map | Filled Map | Circle | Side-by-side Circle



3.6 Worksheets, Dashboards & Stories

Tableau organizes your work into three levels — accessed via tabs at the bottom:



Level	What It Is	Real-Life Equivalent
 Worksheet	Single view with one chart/visualization	One slide with one graph
 Dashboard	Canvas with multiple worksheets combined	Full dashboard page with 4-5 charts
 Story	Sequence of dashboards/sheets as a narrative	PowerPoint presentation with data

💡 Workflow in Tableau:

Start with Worksheets → one chart per sheet

Then combine into a Dashboard → layout multiple charts together

Then tell a Story → guide the viewer through insights step by step

Minimum for a professional report = 3-5 Worksheets + 1 Dashboard

4. Session 1 — Practice & Revision

4.1 Quick Recap

Concept	One-Line Summary
Tableau	Drag-and-drop BI tool for creating interactive data visualizations
Tableau Public	FREE version — perfect for learning, publishes to web
Tableau vs Power BI	Tableau = better visuals Power BI = better Microsoft integration
Dimensions	Categorical fields (blue) — e.g. City, Product, Date
Measures	Numeric fields (green) — e.g. Sales, Profit, Quantity
Columns Shelf	Controls X-axis — drag fields for horizontal layout
Rows Shelf	Controls Y-axis — drag fields for vertical layout
Marks Card	Controls visual properties — Color, Size, Label, Shape, Tooltip
Filters Shelf	Limits which data appears in the view
Show Me Panel	Smart recommender — suggests best chart for selected fields
Worksheet	Single chart view — one visualization per sheet
Dashboard	Multiple worksheets combined in one layout
Story	Sequence of dashboards/sheets as a data narrative

4.2 Practice Questions

Answer the following in your own words:

7. What is Tableau? How is it different from Excel?
8. What is Tableau Public and why will we use it in this course?
9. What are the 5 products in the Tableau family? Name them and their purpose.
10. What is the difference between a Dimension and a Measure in Tableau? Give 2 examples of each.
11. Explain the purpose of Rows Shelf, Columns Shelf and Marks Card in Tableau workspace.
12. What is the difference between a Worksheet, Dashboard and Story in Tableau?

4.3 Hands-on Activity 🖐️

🎯 Explore Tableau Public Interface:

After installing Tableau Public, do the following:

1. Open Tableau Public → Click 'Sample Superstore' from sample data
2. Identify all parts of the interface:
 - Data Pane (Dimensions in blue, Measures in green)
 - Rows Shelf, Columns Shelf, Filters Shelf, Marks Card
 - Show Me Panel (top right)
3. Drag 'Category' to Columns shelf
4. Drag 'Sales' to Rows shelf → A bar chart appears! 📊
5. Drag 'Region' to Color in Marks Card → Bars get colored by region
6. Click Show Me → try different chart types!



Congratulations — you just built your first Tableau visualization!